

The Marine Environment

Exercise 1

A sea	This is a large body of water, such as the Red Sea.
An ocean	This is a major body of saline or salt water such as the Pacific or the Atlantic.
A bay	This is an area of water with land on three sides.
An estuary	This is a semi-enclosed coastal body of water with one or more rivers flowing into it. It is connected to the open sea.
A planet	This is a celestial body, in this case our own planet, the planet Earth.
Fishing	This refers to the sport or business of catching fish (here business).
Shipping	Shipping is the business of delivering goods and sometimes passengers by ship.
Tourism	Tourism is the business of providing things for people to do, places for them to stay, etc. while they are on holiday.
Offshore operations	These are technical activities in the sea or under the seabed at a distance from the coast.

Exercise 2

The marine environment refers to	all living and non-living things that exist naturally in the Earth's seas.
An ecosystem is	a natural unit consisting of all plants, animals and micro-organisms in that unit.
The word " <i>habitat</i> " means	an ecological or environmental area inhabited by a particular animal or plant species.
The IMO aims	to regulate shipping at international level.
MARPOL is aimed at preventing	pollution of the sea by ships.
Climate change is used to refer	to major changes in the weather conditions on Earth, caused by natural events or human activities.
Greenhouse gases are gases	which trap heat above the Earth causing the planet to warm up.

Exercise 3

The marine <**environment**> is a term that refers to all living and non-living things that exist naturally in the Earth's seas.

It contains many fragile <**ecosystems**> such as coral reefs, tropical mangrove forests and island habitats.

Many of these rare ecosystems are internationally under the IMO's <**MARPOL**> convention.

Different ecosystems, including vulnerable marine <**habitats**>, are very important to our planet.

Marine life is often at risk from human activities such as <**fishing**> and <**shipping**>, <**offshore**> operations and <**tourism**>.

<**Climate**> change and <**greenhouse**> gases also <**threaten**> the marine environment.

The Marine Environment and Man-made Pollution

Exercise 1

Pollution in general refers	to the contamination of air, water or soil.
Pollution can be caused	by nature or by man.
We can also categorise pollution	as light pollution, noise pollution, and thermal pollution.
Thermal pollution refers to heat from hot water discharged	by industry into rivers, lakes or seas.
In the texts we will concentrate	on marine pollution.

Exercise 2



Clearly visible marine pollution such as debris in a river



Waste water in a septic river



Ballast water pumped over the side of a cargo ship



A beach covered with oil after an oil spill from a ship



A swan in her nest made of plastic and other garbage



A polluted lagoon



Metals and slowly degrading chemicals form contaminated run-off and sediment



Dead oiled otter, probable victim of an oil spill



Crude oil from a tanker that ran aground



Scientists collecting samples after an oil spill

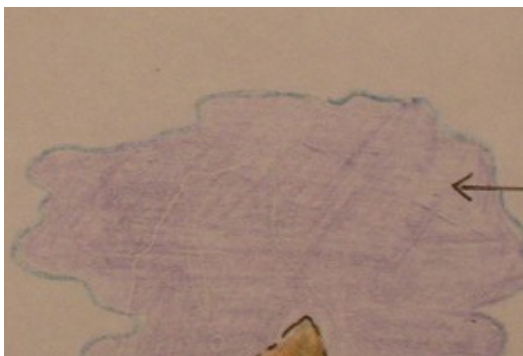
Exercise 3

Pollution is	damage caused to water, air, etc. by harmful substances or waste.
Manmade refers to	something made by humans rather than occurring in nature.
Debris means	wreckage.
Septic describes	a state containing or resulting from disease-causing organisms.
A lagoon is	a shallow body of water, especially one separated from a sea by sandbars or coral reefs.
A sewage pipe	transports water mixed with waste matter.
Ballast water	is used to stabilize a ship.
A victim	is a person who suffers from a harmful act, circumstance, agency or condition.
A sample	is a specimen used for research.
Garbage is	rubbish.

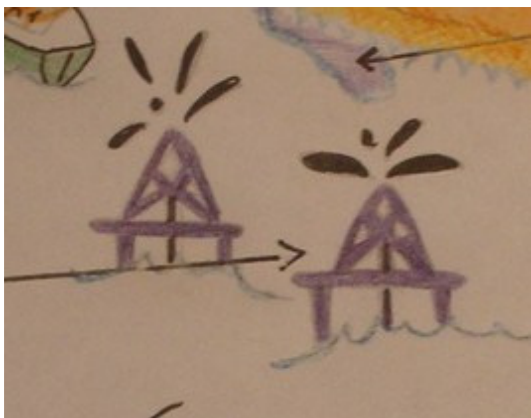
Oil Spills

Exercise 1

An oil slick:



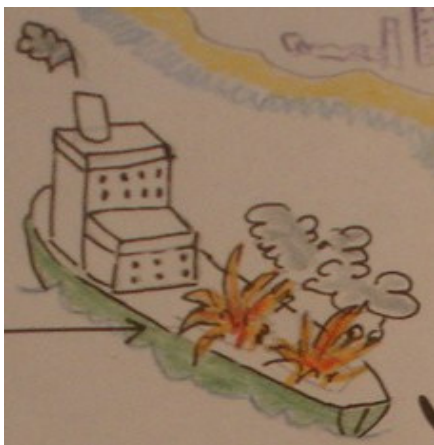
The offshore oil production facility:



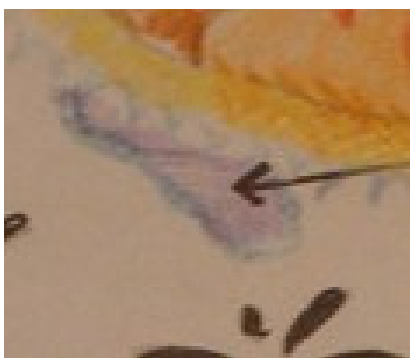
The grounding:



Fire in the cargo holds:



An oil seep:



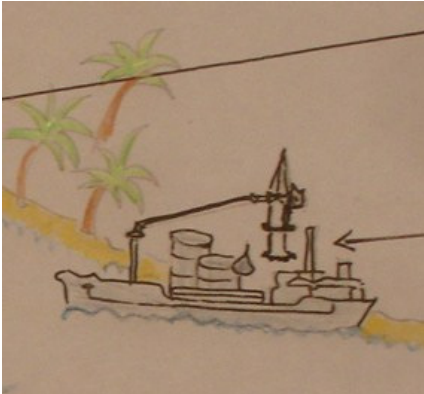
The volcano:



The collision:



The bunkering operations:



Oil Clearance Operations

Exercise 1

The master handed over the vessel to **the third mate**.

The ship spilled 10.8 million gallons of **oil** into the Prince William Sound.

The ship was in the **outbound** shipping lane.

This is a photograph of the **oil tanker**, the Exxon Valdez.

The ship struck **a reef**.

On March 24, 1989, the Exxon Valdez left **Alaska**.

He was assisted by an able seaman, who was **at the helm**.

The mate was trying to avoid **icebergs**.

Exercise 2

Vessel spraying:



Burning:



A boom:



A skimmer:



High pressure hoses:

